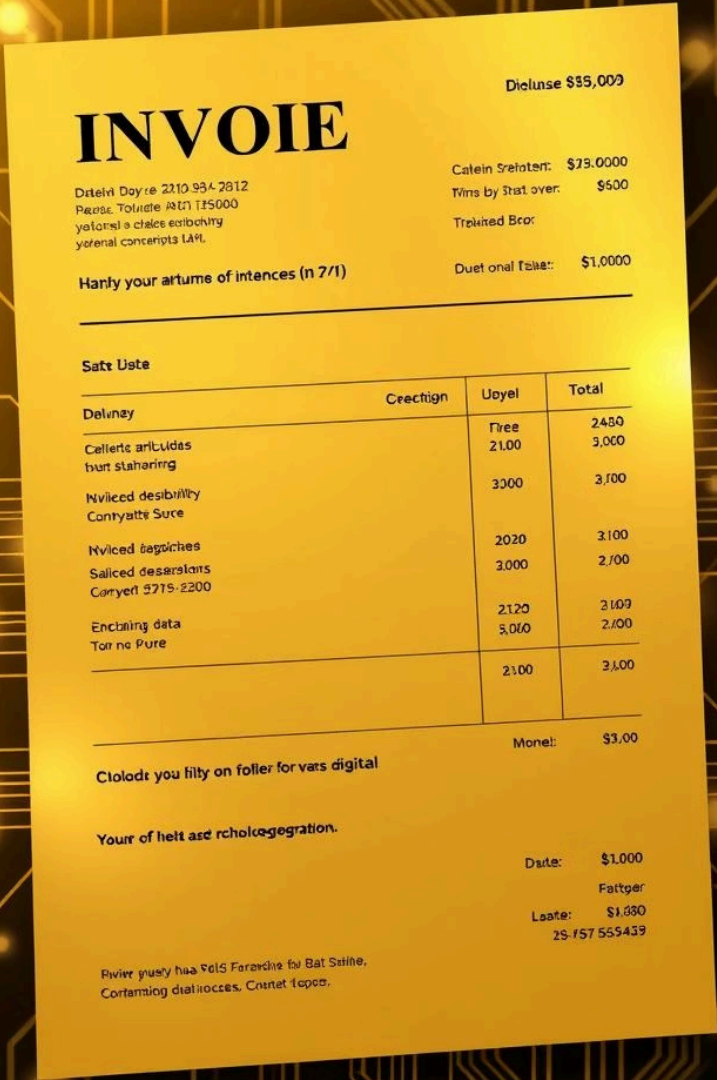


FinExtractAI: Intelligent Invoice Information Extractor

Turning raw invoices into structured insights using Large Language Models (LLMs) and Optical Character Recognition (OCR).



The Challenge: Unstructured Invoice Data



Volume & Variety

Businesses process thousands of invoices daily in diverse, unstructured formats like scanned images and PDFs.



Manual Inefficiency

Traditional manual data entry is time-consuming, prone to human error, and a significant operational expense.



Critical Data Extraction

Extracting key structured data (seller, buyer, date, total) is crucial for financial processes but technically challenging.

FinExtractAI: Solution Overview

FinExtractAI automates invoice data extraction using a sophisticated multi-agent pipeline.



OCR Processing

Initial text extraction from various invoice formats using Tesseract.



LLM Analysis

Semantic entity recognition and data normalisation powered by Gemini Flash and Mistral 7B.



Validation Logic

Rigorous data validation, including date parsing, currency conversion, and consistency checks.



Evaluation & Visualisation

A/B evaluation of LLM performance and intuitive data visualisation for insights.

Modular Agent Architecture

The FinExtractAI system comprises distinct, interoperable agents orchestrated in a unified pipeline.

- 1** **OCRAgent**
Responsible for converting image invoice content into raw, searchable text.
- 2** **SemanticEntityAgent**
Leverages Gemini and Mistral LLMs to identify and extract specific invoice fields, such as seller, invoice number, and amounts.
- 3** **ValidationAgent**
Applies business logic for data integrity, including date format verification and currency exchange rate application.
- 4** **SQLiteAgent**
Manages the persistent storage of extracted and validated invoice data within a SQLite database.
- 5** **VisualiserAgent**
Annotates the original invoice images with the extracted data, providing a visual confirmation of the extraction process.

Core Technologies Utilised

A robust tech stack underpins FinExtractAI's capabilities, ensuring precision and scalability.

- **Programming:** Python
- **OCR:** Tesseract
- **LLMs:** Gemini Flash API, Mistral 7B (Hugging Face)
- **Data Validation:** Custom logic (date parsing, currency conversion)
- **Database:** SQLite
- **Visualisation:** Matplotlib / OpenCV
- **Evaluation Metrics:** Accuracy, Levenshtein Distance, Jaccard Index, F1 Score
- **Deployment:** Streamlit
- **Version Control:** Git

Gemini vs Mistral: Evaluation Snapshot

FinExtractAI leverages leading-edge large language models (LLMs) to extract structured data from complex invoices - reducing manual effort and increasing reliability.

Unmatched Precision

Gemini achieved a perfect 100% accuracy in extracting critical currency information - a true game-changer for financial automation.

Consistent Excellence

Across the board, Gemini's F1-scores exceeded 0.96, showcasing its exceptional and dependable performance.

Fewer Errors, More Confidence

Gemini's lower average edit distance translates to fewer prediction errors and higher confidence.

1 Accurate Seller & Buyer Extraction

2 Unrivalled Currency Identification

3 Precise Date and Total Capture

▼ In Summary

Gemini Flash has proven to be the superior choice, delivering more accurate and reliable extraction of invoice data compared to Mistral 7B. This enabling new levels of automation confidence and efficiency.

Sample Extracted Output

A clear example of structured data retrieved from an unstructured invoice.

```
{  
  "seller": "Lopez, Miller and Romero",  
  "invoice_no": "802205",  
  "invoice_date": "05/08/2007",  
  "buyer": "Hayes LLC",  
  "currency": "USD",  
  "total": "534.11",  
  "Total_in_INR": "46011.10"  
}
```

This JSON format provides readily consumable data for downstream financial systems.

Significant Business Impact

FinExtractAI delivers tangible benefits across financial operations and system integration.



Time Savings

Achieves an 80% reduction in manual data entry, freeing up valuable human resources for strategic tasks.



Accuracy Improvement

Significantly lowers error rates in financial data processing, enhancing data integrity and compliance.



Seamless Integration

Designed for straightforward integration with existing Enterprise Resource Planning (ERP) systems.



Scalable & Flexible

A highly scalable and model-agnostic pipeline, adaptable to future LLM advancements and increasing data volumes.

Key Learnings

Insights gained from the development and evaluation of FinExtractAI.

- **Prompt Engineering:** Mastering effective prompt design for Gemini and Mistral LLMs.
- **Modular ML Systems:** Principles of building robust, extensible machine learning architectures.
- **LLM Orchestration:** Strategies for coordinating LLMs in complex, structured data extraction tasks.
- **MLOps Fundamentals:** Practical application of logging, debugging, and validation in an ML pipeline.
- **OCR + NLP Fusion:** Hands-on experience combining optical character recognition with natural language processing.

Thank You

For your time and attention.

